# Risk-Based Segmentation for Financial Asset Recommendation: Balancing Return, Accuracy, and Risk Suitability

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## Abstract

We present risk-based segmentation, a financial asset recommendation (FAR) approach that routes each customer into a per-band sub-model of LightGCN so that their declared Markets in Financial Instruments Directive II (MiFID II) risk band shapes recommendations during training. On top of this segmentation we ask where suitability is best enforced and compare two exclusive interventions: steering the model during training with a coherence margin loss, or making the data coherent in the first place by regrouping each customer onto the band implied by their purchases. This places risk suitability inside the model, whereas existing FAR systems optimise only return and ranking accuracy or correct for risk by post-hoc re-ranking. We evaluate on three axes, return on investment (ROI), ranking accuracy (nDCG), and risk suitability, which we make measurable with Profile Coherence at $k$ (PC@$k$), and summarise the three with their harmonic mean as a single balance score. An analytical study of FAR-Trans [Sanz-Cruzado *et al.*, 2024] motivates this axis. We find that 18.6% of scoreable Buys lie two or more bands from the customer's declared tolerance, and this discordance is a persistent customer-level trait. A regression with cluster-robust standard errors then shows coherent Buys earn $+2.94$ percentage points higher 6-month realised return than discordant ones, so suitability is not a return tax. Across 69 temporal evaluation splits, the margin loss lifts PC@10 from 0.794 to 0.918 but trades away some accuracy, whereas regrouping lifts PC@10 to 0.949 while also improving nDCG and leaving ROI essentially unchanged, making it the only one of the two interventions to raise the overall balance. Risk suitability can therefore be optimised directly inside a recommender rather than bolted on afterwards, balanced against return and accuracy, offering a template for suitability-constrained recommendation in other regulated domains.

## 1 Introduction

Financial asset recommendation (FAR) identifies and ranks financial securities such as stocks, bonds, and mutual funds for investors according to their suitability. It draws on heterogeneous signals: a customer's past transactions and declared risk tolerance on the investor side, and asset prices, returns, and volatility on the market side. FAR therefore sits at the intersection of personalised ranking and regulatory obligation. The Markets in Financial Instruments Directive II (MiFID II) is a foundational European Union regulation that requires firms to assess suitability before recommending a product, so that the product's risk aligns with the customer's declared risk tolerance and loss capacity. A recommender that maximises predicted return or ranking quality without conditioning on this declared risk band can excel on standard benchmarks yet remain a regulatory liability.

FAR-Trans [Sanz-Cruzado *et al.*, 2024] is the first open FAR benchmark to pair real asset pricing with real retail-investor transactions and a declared MiFID II risk band per customer, collected from a large European financial institution over January 2018 to November 2022. Its accompanying benchmark evaluates FAR algorithms on two axes: ranking accuracy, via normalised Discounted Cumulative Gain (nDCG@10), and realised profitability, via Return on Investment (ROI@10). A subsequent study finds these two axes are themselves negatively correlated on FAR-Trans, so a model that wins on accuracy tends to lose on return [Sanz-Cruzado *et al.*, 2026].

Neither axis measures whether a recommendation is suitable for the customer's risk band. A model can lead the nDCG or ROI leaderboard while routinely proposing assets two or more bands away from the customer's declared tolerance, the precise failure MiFID II exists to prevent. Risk-aware FAR methods address this only after the fact, re-ranking a base model's top list by a risk-tolerance utility [Sakurai *et al.*, 2026] rather than conditioning the model on the band. The declared risk band therefore never enters training, and risk suitability is left to a post-hoc filter or ignored outright.

Our objective is to bring risk suitability, the alignment between an asset's risk and the customer's declared band, inside the FAR model and to balance it against return and accuracy rather than trade one for another. We evaluate every model on three axes: return (ROI@10), ranking accuracy (nDCG@10), and risk suitability, which we make measurable by introducing Profile Coherence at $k$ (PC@$k$). PC@$k$ is the share of a customer's top-$k$ list whose asset risk class lies within one band of the customer's declared MiFID II band, on a four-point ordinal scale from Conservative to Aggressive. To capture how well a recommender reconciles the three, we further introduce a single balance score, their harmonic mean, which is high only when return, accuracy, and risk suitability are strong together and falls sharply when any one is sacrificed. PC@$k$ and this balance are themselves new, but our central contribution is a recommender that achieves the balance they measure.

That recommender is risk-based segmentation of LightGCN [He *et al.*, 2020]: we partition customers by declared band and route each into a dedicated sub-model during training, so the band shapes the learned embeddings directly rather than filtering its output. On top of this segmentation baseline we ask where risk suitability is best enforced and study two exclusive interventions: steering the model during training with a coherence margin loss atop Bayesian Personalised Ranking (BPR) [Rendle *et al.*, 2012] that penalises scoring a discordant asset above a coherent one, or making the data coherent in the first place by regrouping each customer onto the band implied by their revealed purchases.

We make three contributions:

- **An analytical study of risk factors in FAR-Trans.** Of 228,241 scoreable Buy transactions, 18.6% land two or more bands from the customer's declared tolerance, and this discordance is a persistent customer-level trait concentrated at the Conservative and Aggressive extremes rather than transaction-level noise. A transaction-level regression on 208,029 priced Buys, with cluster-robust standard errors, shows coherent Buys earn $+2.94$ percentage points higher 6-month realised return than discordant ones, conditional on volatility, segment, and year. This is evidence that risk suitability carries no return penalty.
- **A risk-based segmentation approach for LightGCN.** We route each customer to a per-band sub-model so that the declared risk band conditions the learned embeddings, the baseline methodology on which the experiments build.
- **Experimental studies across the three axes.** Over 69 temporal evaluation splits we audit the strongest FAR-Trans baselines, LightGCN (best accuracy) and Random Forest (best return), and ablate two exclusive interventions on the segmentation baseline: steering the model with a coherence margin loss, and making the data coherent through regrouping. The baselines under-serve specific risk bands despite adequate aggregate scores. The margin loss lifts PC@10 from 0.794 to 0.918 but trades away some accuracy, whereas regrouping lifts PC@10 to 0.949 while also improving nDCG and leaving ROI essentially unchanged, so of the two levers only regrouping raises the harmonic-mean balance of the three axes.

# 2 Preliminaries

This section sets up the financial asset recommendation problem and the FAR-Trans benchmark that grounds our study, then the two-axis evaluation standard whose blind spot motivates the third axis this paper adds.

## 2.1 The Financial Asset Recommendation Problem

Financial asset recommendation (FAR) identifies and ranks financial securities such as stocks, bonds, and mutual funds for an investor according to their suitability. Suitability is two-sided. Investor-side signals include the customer's past transactions, declared risk tolerance, investment capacity, and personal goals, while market-side signals include asset returns, volatility, currency value, and inflation. A FAR system therefore draws on heterogeneous data sources: time-series pricing data, customer profile data, and historical investment transactions. Crucially, the investor-side half of suitability is a regulatory requirement under MiFID II, so a recommendation is only as good as its fit to the customer's risk profile, not its expected return or ranking quality alone.

## 2.2 The FAR-Trans Benchmark

Most prior FAR models were developed over proprietary or simulated data, which prevents fair cross-method comparison, and the only earlier public dataset lacked pricing data and asset identifiers, ruling out price-based approaches. FAR-Trans [Sanz-Cruzado *et al.*, 2024] closes this gap. It is the first public FAR dataset that pairs real asset pricing with real retail-investor transactions, collected from a large European financial institution and spanning January 2018 to November 2022. It provides daily closing prices, 228,913 Buy and 159,135 Sell transactions over 806 assets and 29,090 predominantly retail customers, and customer profile metadata including a declared MiFID II risk band per customer. Together these make both price-based and profile-based recommendation measurable on a single dataset.

Alongside the data, FAR-Trans establishes the field's evaluation standard by benchmarking 11 FAR algorithms. Two anchor our study because each leads on one axis of that benchmark: LightGCN [He *et al.*, 2020], a graph-based collaborative filter that propagates embeddings over the user-item bipartite graph and is the strongest ranking model, and Random Forest, a tree-based ensemble that ranks assets from tabular customer and asset features and is the strongest profitability model. The benchmark scores both over the top-10 ranked items on two axes: normalised Discounted Cumulative Gain (nDCG@10) for next-purchase ranking accuracy and Return on Investment (ROI@10) for realised profitability.

## 2.3 The Suitability Blind Spot

These two axes are not aligned on FAR-Trans. A subsequent study [Sanz-Cruzado *et al.*, 2026] shows that transaction-based and profitability-based metrics are negatively correlated, so a model that wins on one tends to lose on the other, and concludes that ranking accuracy and realised return capture genuinely different objectives. Yet neither axis measures whether a recommendation aligns with the customer's declared MiFID II risk band. That regulatory suitability dimension is the blind spot this paper targets: we treat it as a third evaluation axis, risk suitability, and ask whether it can be balanced against return and accuracy rather than traded off. The next section makes risk suitability measurable and establishes that it is both prevalent enough to matter and economically consequential.

# 3 Analytical Study of Risk Factors

This section establishes the risk-suitability axis empirically on FAR-Trans. It first introduces the machinery the rest of the paper relies on: the ordinal risk bands shared by customers and assets, the rule that assigns each asset a band, and the Profile Coherence metric that scores risk suitability. It then asks two questions of the observed Buy transactions. First, how prevalent is discordance, and is it a stable customer trait or transaction-level noise? Second, do discordant Buys earn lower realised 6-month return than coherent ones once asset volatility, customer segment, and year are controlled for?

## 3.1 Risk Bands

### Ordinal band scale

Customers and assets share the same 4-point ordinal scale running from Conservative to Income to Balanced to Aggressive, encoded 0 to 3. The customer band $b_u$ is taken from FAR-Trans's `riskLevel` field. FAR-Trans provides both declared values and regression-imputed `Predicted_*` entries, and both are mapped to the same ordinal scale. Customers whose entry is `Not_Available` or missing are excluded from all coherence computations. The asset band $b_i$ is assigned by the three-step hierarchical rule below, applied in order.

### Step 1: subcategory metadata

Mutual funds are labelled directly from their subcategory: Money Market funds are treated as Conservative, Bond funds as Income, Balanced as Balanced, and Equity or Large Cap as Aggressive. Bond instruments are labelled by issuer type: Government as Conservative, Corporate as Income, and any remaining bond subcategory as Income by default.

### Step 2: volatility quartiles for stocks

Stocks carry no subcategory tag we can use, so they fall through to a volatility rule. For each ISIN we compute the annualised standard deviation of daily log returns on a trailing 252-day window, then bucket all stocks into the four bands by the stock-only quartile thresholds $(q_1, q_2, q_3)$.

### Step 3: residual default

Any asset that falls through both prior steps is assigned to Balanced.

Applying the three steps above produces an asset-band distribution of $(190, 333, 105, 178)$ across (Conservative, Income, Balanced, Aggressive) over $|A| = 806$ assets.

## 3.2 Profile Coherence

Risk suitability is the alignment between an asset's risk band and the customer's. We make it measurable with Profile Coherence, built from the ordinal distance between the two bands.

### Pairwise discordance

$$d(u, i) = |b_u - b_i| \in \{0, 1, 2, 3\}. \tag{1}$$

A recommendation is *profile-coherent* iff $d \le 1$. We adopt this $\pm 1$ band tolerance by design, to mirror MiFID II's adjacency allowance for borderline suitability rather than to penalise every single-band drift as a violation.

### Aggregation

$$\text{PC@}k(u) = \frac{1}{k}\left|\{i \in \text{top}_k(u) : d(u, i) \le 1\}\right|. \tag{2}$$

Two implementation rules complete the definition. First, recommenders returning fewer than $k$ items use the actual number of items returned as the denominator. Second, customers with no assigned risk band ($b_u = \emptyset$) are dropped from the aggregate. Per-split PC@$k$ is the unweighted mean across eligible customers and the headline numerics average over all splits.

## 3.3 Dataset

Beyond the dataset scale given in Section 2, two profile details matter for coherence. Of the 29,090 customers, 28,770 (98.9%) carry a declared or regression-imputed MiFID II risk band, and the remaining 320 are excluded from all coherence computations. Customer profiles also include `customerType`, which enters our regression study later as a control.

## 3.4 Prevalence and Structure of Discordance

We characterise discordance in four slices, starting from the pooled transaction rate and drilling into the customer, band, and year dimensions in turn. At the transaction level (Figure 1), of 228,241 scoreable Buys:

1. 18.6% violate the declared band by two or more steps.
2. 81.4% are coherent, and 34.1% are strictly coherent ($d = 0$).
3. Mean discordance is 0.874 bands across all scoreable Buys, indicating that when mismatches occur they are usually around a single band step.

All in all, nearly one in five scoreable Buys lands two or more bands away from the customer's declared tolerance. That violation rate is too large to dismiss as sampling noise, and it is the empirical motivation for treating risk suitability as a measurable evaluation axis rather than a property the baselines can be assumed to satisfy.

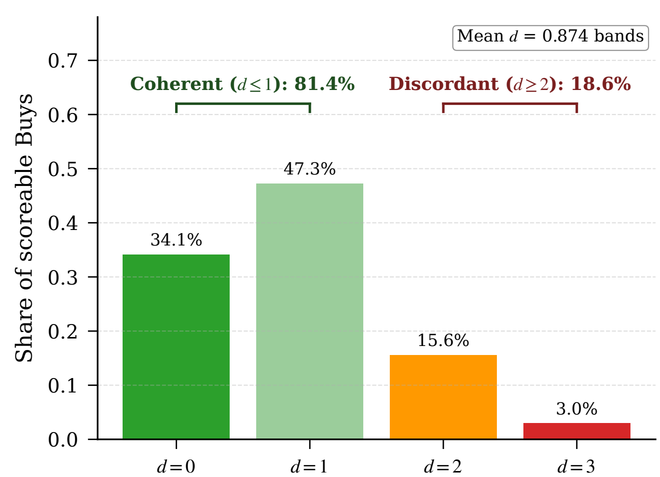


Figure 1: Distribution of pairwise discordance $d$ over 228,241 scoreable Buys. Most mass sits at $d \le 1$ (coherent), with non-negligible density at $d \ge 2$ (discordant).

The pooled rate cannot tell us whether these violations are i.i.d. transaction-level slips spread thinly across the customer base or a structural trait concentrated in a subset. Slicing by customer settles the question. The per-customer discordant share (Figure 2) is sharply bimodal, concentrating in the lowest and highest bins rather than clustering near the 18.6% population mean. More explicitly, 64.4% of banded customers are fully coherent across every Buy and 17.2% are fully discordant.

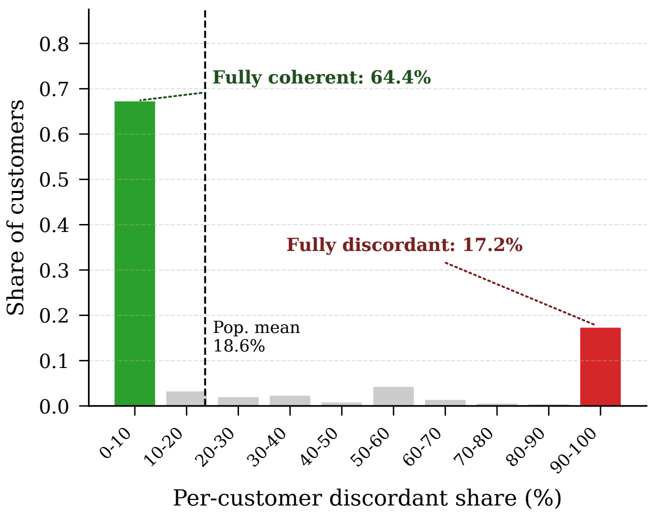


Figure 2: Per-customer share of discordant Buys is sharply bimodal, concentrating in the lowest and highest bins and ruling out i.i.d. transaction-level noise.

Hartigan's dip test on the 28,770 per-customer values rejects unimodality ($D = 0.086$, $p < 10^{-15}$), confirming that discordance is a persistent customer-level trait rather than i.i.d. transaction-level noise.

Since the trait is customer-level, the next question is which customers carry it. Coherence per declared band is U-shaped (Figure 3). The two centre bands sit around 90%, while Conservative drops to 45.1% (possibly reflecting a "reach for yield" tendency) and Aggressive to 56.2% (possibly reflecting a regression towards safer assets). This concentration of mismatch at the ordinal extremes corroborates the bimodality finding from the customer slice.

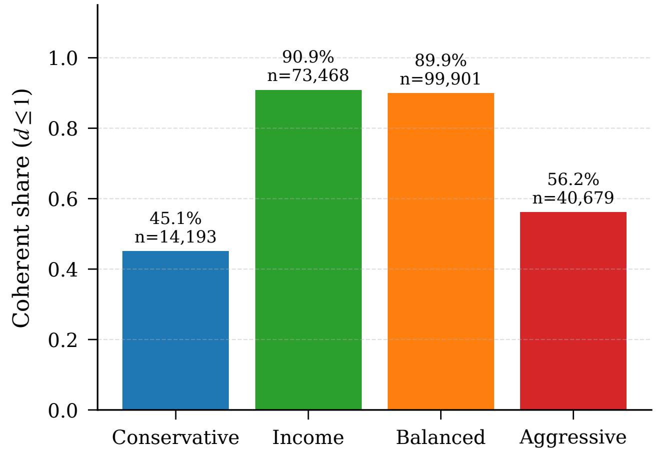


Figure 3: Coherence share by declared band. The U-shape supports the bimodality result of Figure 2: discordance concentrates on the ordinal extremes.

A final slice rules out a regime-driven artefact. Mean discordance is flat across 2019 to 2022 (within 0.02 bands, see Figure 4), so the pattern is not the product of any single year's market conditions. The 2018 value (0.98) sits higher. MiFID II came into force on 2018-01-03, the exact start of FAR-Trans, so we hypothesise that the elevation is a transition effect. However, this hypothesis is not tested here and is left to future work.

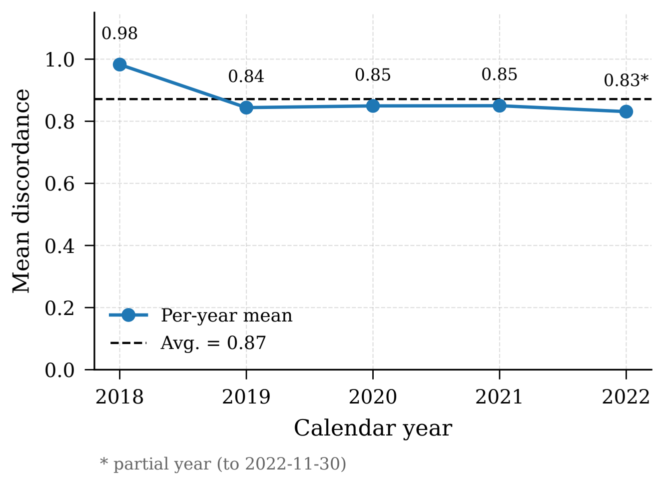


Figure 4: Mean discordance by calendar year. The 2019 to 2022 cluster sits within 0.02 bands, with 2018 elevated (transition-effect hypothesis).

### 3.5 Conditional Return Premium

We have shown that discordance is a persistent customer-level trait concentrated at the ordinal extremes. The next question is whether it carries an economic cost: do discordant Buys earn lower returns than coherent ones?

**Regression specification**

We test whether profile-coherent Buys earn different realised returns from discordant ones via a transaction-level Ordinary

| Term | Estimate | Std. Error | 95% CI | $p$ |
|---|---|---|---|---|
| Intercept | $-0.0471$ | 0.0141 | $[-0.0746, -0.0195]$ | $8.1 \times 10^{-4}$ |
| `is_coherent` | $+0.0294$ | 0.0040 | $[+0.0215, +0.0372]$ | $2.3 \times 10^{-13}$ |
| `asset_volatility` | $-0.1025$ | 0.0165 | $[-0.1349, -0.0701]$ | $5.5 \times 10^{-10}$ |
| `C(customer_type)[Legal Entity]` | $+0.0154$ | 0.0486 | $[-0.0798, +0.1106]$ | 0.751 |
| `C(customer_type)[Mass]` | $+0.0011$ | 0.0132 | $[-0.0248, +0.0269]$ | 0.936 |
| `C(customer_type)[Premium]` | $+0.0058$ | 0.0135 | $[-0.0206, +0.0322]$ | 0.667 |
| `C(customer_type)[Professional]` | $+0.0247$ | 0.0169 | $[-0.0084, +0.0578]$ | 0.143 |
| `C(year)[2019]` | $+0.1759$ | 0.0129 | $[+0.1507, +0.2012]$ | $1.4 \times 10^{-42}$ |
| `C(year)[2020]` | $+0.2107$ | 0.0061 | $[+0.1986, +0.2227]$ | $6.2 \times 10^{-259}$ |
| `C(year)[2021]` | $+0.0754$ | 0.0053 | $[+0.0651, +0.0858]$ | $4.1 \times 10^{-46}$ |
| `C(year)[2022]` | $-0.0195$ | 0.0045 | $[-0.0283, -0.0107]$ | $1.4 \times 10^{-5}$ |

Table 1: Full OLS coefficient table for the return-premium regression on 208,029 priced Buys, cluster-robust SE on `customerID`. Reference cell: Inactive `customerType` (the alphabetically first segment in FAR-Trans), year 2018, `asset_volatility` $= 0$, `is_coherent` $= 0$.

Least Squares (OLS) regression on 208,029 priced Buys with cluster-robust standard errors on `customerID` and a two-sided Wald test at $\alpha = 0.05$. Clustering is warranted because each customer contributes roughly 8 Buys on average, so within-customer residuals share unobserved shocks. The specification is:

$$
\begin{aligned}
r = \beta_0 &+ \beta_1 \cdot \text{is\_coherent} \\
&+ \beta_2 \cdot \text{asset\_volatility} \\
&+ \mathbf{C}(\text{type})\boldsymbol{\gamma} \\
&+ \mathbf{C}(\text{year})\boldsymbol{\delta} \\
&+ \varepsilon,
\end{aligned}
\tag{3}
$$

where $r = (p_{\text{end}} - p_{\text{start}})/p_{\text{start}}$ is the 6-month realised return as a decimal fraction (e.g., $r = 0.03$ means a 3 percentage-point gain) and `is_coherent` is binary ($d \leq 1$).

Prices are matched via a backward-nearest join: the start price is the most recent close on or before the trade date, and the end price is the most recent close on or before the date six months later. Transactions without a price match within 7 days at either endpoint are dropped. The coefficient of interest is $\beta_1$ on `is_coherent`, and the null hypothesis $H_0 : \beta_1 = 0$ states that coherence has no conditional effect on 6-month returns.

**Results**

Table 1 reports the full coefficient table. The key results on the coherence premium are:

- **Conditional premium.** Coherent Buys earn $+2.94$ percentage points (pp) higher 6-month realised return than discordant ones, conditional on volatility, segment, and year.
- **Significance.** The positive sign with $p = 2.3 \times 10^{-13}$ provides strong evidence against the prior that regulatory coherence imposes a return tax.

We note that the adjusted $R^2$ is 0.051, which implies that the covariates explain little of the variation in individual returns. This is expected when idiosyncratic price movements dominate, which concerns prediction but does not affect our hypothesis test. The coherence premium is an average difference in return between coherent and discordant Buys, and even with a low $R^2$ it is estimated precisely. We cluster the standard errors by `customerID` so that repeated Buys from the same customer are not counted as independent evidence.

Turning to the controls, the volatility coefficient is $-10.25$ pp per unit, consistent with low-volatility outperformance over 2019 to 2022 and the 2022 rate shock that hit high-beta names. All four year dummies reject $H_0$, reflecting distinct market regimes relative to the 2018 reference year:

1. The 2019 coefficient reflects the broad equity recovery in the year after the December 2018 sell-off, which had left the S&P 500 just short of bear-market territory [CNBC, 2019].
2. 2020 is the strongest at $+21.07$ pp, coinciding with the rapid rebound from the COVID-19 sell-off that returned major indices to record highs by August [Reuters, 2020].
3. 2021 reflects continued gains under still-accommodative monetary policy, before the Federal Reserve's expected late-2021 taper and rising rate-hike expectations [Reuters, 2021].
4. The 2022 coefficient turns negative at $-1.95$ pp, the year an aggressive Federal Reserve hiking cycle drove a broad fixed-income drawdown [Reuters, 2022].

Lastly, all four `customerType` dummies fail to reject $H_0$, indicating that segmentation contributes no detectable return signal. Together with the bimodality found above, this suggests that customer type does not capture the persistent discordance trait either.

Taken together, these controls barely move the premium: the raw slice gap is $+3.31$ pp (unconditional mean 6-month return: 4.23% coherent vs 0.92% discordant), whereas the `is_coherent` coefficient, which already conditions on volatility, segment, and year, is $+2.94$ pp. Adding the controls therefore absorbs only 0.37 pp ($\sim$11% of the unconditional gap), so coherence is not a volatility, segment, or year confound. The coherence premium therefore survives all controls we can measure, motivating the question of whether existing recommenders produce coherent lists at all.

### 3.6 Limitations

Two caveats bound the findings of this section.

- *Observational nature.* The coherence premium is associational, as unobserved confounders such as financial literacy, advisor channel, account constraints, and prior portfolio composition are not ruled out by the volatility, segment, or year controls. Similarly, the elevated 2018 mean discordance is hypothesised as a MiFID II transition effect but not directly tested against a pre-MiFID II counterfactual.
- *Asset-band assignment.* The asset bands $b_i$ are not provided by FAR-Trans but assigned by the hierarchical heuristic in Section 3.1 rather than an externally validated classification, so any systematic mislabelling, especially from the volatility-quartile step for stocks, propagates into all coherence numbers.

# 4 Methodology: Risk-Based Segmentation

Section 3 established that discordance is prevalent, structural, and economically consequential, yet the FAR-Trans baselines optimise only return and ranking accuracy. Our methodology brings risk suitability inside the recommender. The core idea is *risk-based segmentation*: instead of a single global model, we route each customer to a sub-model trained only on customers of their risk band. Figure 5 gives the full picture, from the discordance challenge through the segmentation backbone and its two alternatives to the harmonic-mean balance that ties the three axes together. We describe the baseline it improves on, the segmentation architecture, the two alternatives for raising risk suitability, and the balance score in turn.

## 4.1 From a Global Model to Per-Band Segmentation

The FAR-Trans LightGCN baseline is a single collaborative filter over one user-item bipartite graph that connects every customer to the assets they bought. It learns an embedding per customer and per asset, propagates them over the graph, and scores an asset for a customer by the inner product of their embeddings, training with the Bayesian Personalised Ranking (BPR) objective [Rendle *et al.*, 2012]. The model never sees the customer's risk band, so suitability is whatever the purchase graph happens to imply. Because Section 3 showed that purchases are frequently discordant, a band-agnostic model trained to reproduce them inherits that discordance.

## 4.2 Risk-Based Segmentation

We choose LightGCN as the backbone because its differentiable loss and per-customer embeddings let us condition on the risk band and, later, attach a coherence margin directly to the BPR objective. Prior work shows LightGCN readily carries domain-specific structure beyond plain BPR [Ghiye *et al.*, 2024]. Random Forest, the other FAR-Trans anchor, ranks assets from technical indicators alone and carries no customer-level signal to segment on, so we do not extend it.

Risk-based segmentation partitions the customer base by declared MiFID II band and trains one LightGCN sub-model per band (Conservative, Income, Balanced, Aggressive). Each sub-model's training graph contains only the interactions of customers in that band, so its embeddings specialise to that band's preferences. A band router directs each customer at both training and inference: a customer is served by the sub-model of their declared band, and customers with no declared band fall back to the Balanced sub-model. Each sub-model is trained with the same BPR objective and edge-dropout regularisation as the baseline, changing only the population it learns from.

## 4.3 Two Alternatives for Raising Risk Suitability

Segmentation specialises each sub-model to a band but still trains purely to reproduce purchases, which Section 3 showed are themselves often discordant. We therefore study two exclusive alternatives for pushing risk suitability higher, one acting on the model and one on the data. Their mechanics are introduced here, and the experiments evaluate them.

**Alternative 1: steering the model (margin loss).** BPR trains the model to score a bought asset (the positive) above a random unobserved asset (the negative). Independently of this sampling, we add a coherence margin that draws, from the sub-model's asset universe, one coherent asset $i_c$ ($d(u, i_c) \leq 1$) and one discordant asset $i_d$ ($d(u, i_d) > 1$), and pushes the coherent score above the discordant one:

$$\mathcal{L}_{\text{coh}}(u) = \text{softplus}(s(u, i_d) - s(u, i_c)), \qquad (4)$$

where $s(u, i)$ is the inner-product score and $\text{softplus}(x) = \ln(1 + e^x)$ smoothly approximates the hinge $\max(0, x)$. The total objective adds this term to BPR with weight $\lambda \geq 0$:

$$\mathcal{L} = \mathcal{L}_{\text{BPR}} + \omega\|\Theta\|_2^2 + \lambda \cdot \mathcal{L}_{\text{coh}}, \qquad (5)$$

where $\omega$ is the L2 weight decay on embedding parameters $\Theta$. BPR remains the primary ranking signal: setting $\lambda = 0$ recovers plain segmentation, while $\lambda > 0$ steers scores towards coherent assets. We use softplus rather than a hard hinge because it is differentiable everywhere and matches the form of BPR's own $-\ln \sigma(\cdot)$ loss, keeping the two objectives consistent.

**Alternative 2: fixing the data (regrouping).** The router and the margin loss both key off the declared band, which Section 3 showed is frequently inconsistent with revealed behaviour. The second alternative corrects the label at its source: before training, each banded customer is reassigned to the band implied by their purchases, taken as the lower median of the risk classes of the assets they bought (even-count ties resolve to the more conservative band). Segmentation then routes and trains on these regrouped bands, making the data coherent in the first place rather than penalising incoherence during training.

## 4.4 Balancing the Three Axes

We evaluate every configuration on the three axes of Section 2: return (ROI@10), ranking accuracy (nDCG@10), and risk suitability (PC@10). Because these axes can trade off, we summarise them with a single balance score, their harmonic mean:

$$\text{Balance} = 3\left(\frac{1}{\text{ROI}'} + \frac{1}{\text{nDCG@10}} + \frac{1}{\text{PC@10}}\right)^{-1}, \qquad (6)$$

where $\text{ROI}'$ denotes ROI@10 rescaled to the same $[0, 1]$ range as the other two axes, the mapping given in Section 5.2. The

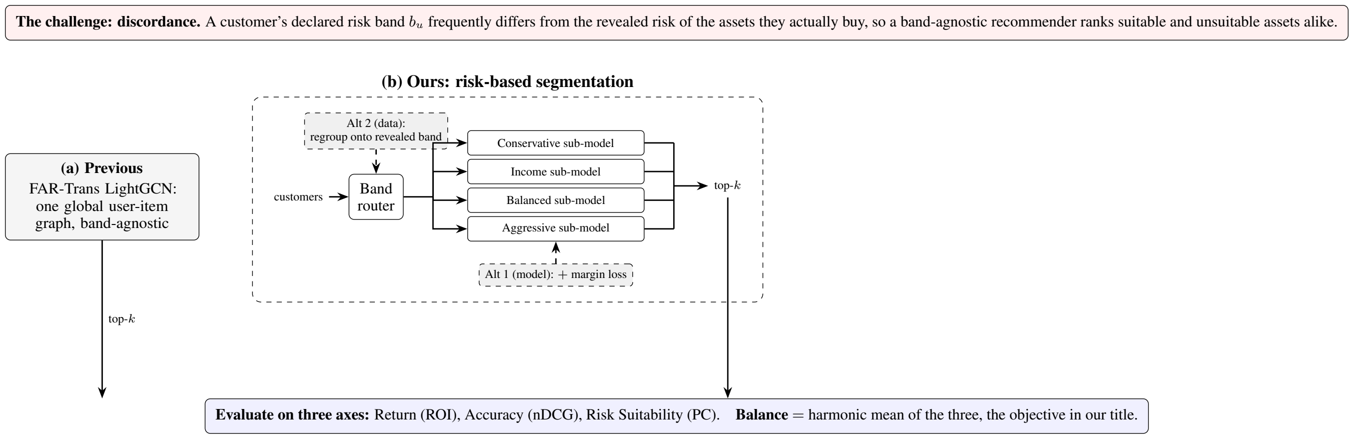


Figure 5: Overview of risk-based segmentation. The previous FAR-Trans LightGCN baseline (a) trains one band-agnostic model over the global user-item graph. Our approach (b) routes each customer by declared MiFID II band to a dedicated LightGCN sub-model, with two exclusive alternatives for raising risk suitability: a model-side margin loss (Alt 1) and a data-side regrouping onto the revealed band (Alt 2). Every configuration is scored on return, accuracy, and risk suitability, summarised by their harmonic-mean balance.

harmonic mean is high only when return, accuracy, and risk suitability are strong together and falls sharply if any one is sacrificed, directly operationalising the balance our title targets.

# 5 Experimental Setup

This section specifies how every configuration is evaluated: the temporal split protocol, the metrics that instantiate the three axes and the balance summary, the baseline hyperparameter grids, and the configurations compared in the ablation. The results follow in the next section.

## 5.1 Temporal Split Protocol

Evaluation uses 69 temporal splits with approximately bi-weekly split points on an evenly spaced trading-day grid running from August 2019 to December 2021, with per-split test windows extending evaluation coverage to May 2022. The two-week split cadence follows the original FAR-Trans benchmark [Sanz-Cruzado *et al.*, 2024]. For each split, the training set contains all Buys in $[0, \texttt{time\_point}]$ and the test set contains all Buys in $[\texttt{time\_point}, \texttt{test\_end}]$, deduplicated against training and filtered to the training-asset universe. Eligible customers have at least one interaction in both training and test, while eligible assets have prices on both endpoints.

## 5.2 Evaluation Metrics

We score every recommender over its top-10 ranked items on the three axes of Section 2, then summarise them with a single balance.

**Return: ROI@10.** For each recommended asset we form the holding-period return from its close price at the split's `time_point` and at `test_end`, convert it to a per-month geometric rate, average over the ten assets, and then over eligible customers. ROI@10 is reported per month.

**Accuracy: nDCG@10.** Ranking accuracy is the rank-discounted hit rate of the held-out test Buys within the top 10, normalised by the ideal ordering and averaged over eligible customers.

**Risk suitability: PC@10.** Profile Coherence is defined in Section 3.2. To read per-band coherence on a common footing, we normalise it by a band-conditional random baseline. The closed-form expectation of PC@$k$ under a uniformly random recommender for band $b$ is

$$\pi(b) = \frac{|\{i \in A : |b - b_i| \leq 1\}|}{|A|}, \tag{7}$$

where $A$ is the asset universe. With the asset-band distribution of Section 3.1 this yields $\pi(b) = (0.65, 0.78, 0.76, 0.35)$ for (Conservative, Income, Balanced, Aggressive). Per-user coherence is then normalised against the random reference at the customer's own band,

$$\text{PC-lift@}k(u) = \frac{\text{PC@}k(u)}{\pi(b_u)}, \tag{8}$$

so that lift $> 1$ beats the band-conditional random rate. Because 77.9% of FAR-Trans assets sit in the Conservative, Income, and Balanced bands, a recommender that recommends Income assets to everyone would score well on raw PC@$k$ for any customer whose tolerance set includes Income. Dividing by $\pi(b_u)$ removes this advantage, so bands that are easier to satisfy by chance are not credited for it and the lift values are comparable across bands.

**Balance.** Section 4 defined the balance as the harmonic mean of the three axes with ROI rescaled. We complete that definition here. nDCG@10 and PC@10 are already rates in $[0, 1]$, but ROI@10 can be negative and lives on a different scale, so we map it onto $[0, 1]$ with a logistic transform

$$\text{ROI}' = \frac{1}{1 + e^{-\text{ROI@10}/\tau}}, \qquad \tau = 0.01, \tag{9}$$

which sends a flat return to $0.5$ and moves smoothly towards $1$ for gains and $0$ for losses, with $\tau$ setting the scale at one percentage point per month. This rescaled ROI$'$ feeds the

| Parameter | Search space | Best | Role |
|---|---|---|---|
| **LightGCN** (tuned on average nDCG@10) | | | |
| `embedding_dim` | $\{64, 128\}$ | 64 | grid |
| `num_layers` | $\{2, 3\}$ | 3 | grid |
| `learning_rate` | $\{10^{-2}, 10^{-3}\}$ | $10^{-3}$ | grid |
| `weight_decay` | $10^{-5}$ | n/a | fixed |
| `keep_prob` | 0.6 | n/a | fixed |
| `num_epochs` | 50 | n/a | fixed |
| `batch_size` | 1024 | n/a | fixed |
| **Random Forest** (tuned on average ROI@10) | | | |
| `num_estimators` | $\{20, 30, 40, 50\}$ | 40 | grid |
| `max_depth` | $\{15, 25, 50\}$ | 50 | grid |
| horizon | 6 months | n/a | fixed |
| `random_state` | 42 | n/a | fixed |

Table 2: Baseline hyperparameter grids and best-trial values, swept with Ray Tune. "grid" rows are searched, "fixed" rows are held constant.

harmonic-mean balance defined in Section 4, which is high only when return, accuracy, and risk suitability are strong together and collapses if any one is sacrificed.

### 5.3 Baselines and Hyperparameter Grids

Table 2 reports the search grids, fixed values, and best-trial selections for the two FAR-Trans baselines. Both grids are swept with Ray Tune, with LightGCN tuned on average nDCG@10 and Random Forest on average ROI@10 across the 69 splits.

### 5.4 Configurations and Per-Band Audit

We compare the following configurations on the same 69 splits, each reported on all three axes and the balance:

- **Baseline.** The FAR-Trans LightGCN at its best trial, and the Random Forest baseline for the per-axis reference.
- **Segmentation.** Risk-based segmentation alone (the coherence margin off), isolating the contribution of the architecture.
- **Model lever.** Segmentation with the coherence margin loss added.
- **Data lever.** Segmentation retrained and routed on the regrouped (revealed) bands.

The model lever and the data lever are the two exclusive interventions of Section 4, each applied on top of the segmentation baseline. All segmented configurations inherit the best LightGCN backbone hyperparameters of Table 2, and the segmentation is trained with the same BPR objective and edge-dropout regularisation as the baseline. We additionally report the combined setting (regrouping with the margin loss) for completeness.

For the per-band audit, we average per-customer PC@10 unweighted over all (customer, split) pairs within each declared band, and report per-band PC-lift@10 against $\pi(b)$. This exposes which bands a recommender actually serves, which the aggregate PC@10 can mask because the random baselines $\pi(b)$ differ widely across bands.

## 6 Results and Discussion

We report the experiments in three steps. We first audit where the FAR-Trans baselines sit on the three axes and across the four declared bands. We then evaluate risk-based segmentation and its two exclusive levers on the aggregate balance, the per-band suitability, and the per-split consistency. We close by explaining why correcting the data beats penalising the model. Aggregate metrics over the 69 splits are in Table 3, per-band suitability in Table 4.

### 6.1 Baseline Audit

The two FAR-Trans baselines occupy opposite corners of the return-accuracy tradeoff and neither balances the three axes (Table 3). Random Forest wins return ($+1.4\%$/mo) but ranks near random (nDCG@10 of $0.019$), so its balance collapses to $0.054$. LightGCN wins ranking accuracy (nDCG@10 of $0.330$, PC@10 of $0.794$) but earns a slightly negative ROI@10, for a balance of $0.428$. Aggregate PC@10 looks adequate for both (LightGCN $1.17\times$ lift, Random Forest $1.06\times$), but that single number averages over bands whose random baselines $\pi(b)$ range from $0.35$ to $0.78$ and so hides which bands are actually served.

Table 4 exposes the gap. For both baselines, per-band lift rises monotonically with the declared band: $0.52 \rightarrow 0.70 \rightarrow 1.03 \rightarrow 1.88$ for Random Forest and $0.75 \rightarrow 1.13 \rightarrow 1.17 \rightarrow 1.35$ for LightGCN. Recommendations skew towards higher-risk assets, so both baselines under-serve the Conservative band (below the random baseline), and Random Forest also under-serves Income. Aggressive's high lift is partly mechanical, since $\pi(\text{Aggressive}) = 0.35$ is the lowest bar to clear. Neither baseline carries band labels or a band-aware objective, so per-band coherence is a passive byproduct of training rather than a controlled output. This is the deficit a band-aware approach must close.

### 6.2 Aggregate Results of Risk-Based Segmentation

Reading the ablation down Table 3 separates a cheap architectural step from the two levers. Segmentation alone lifts PC@10 to $0.821$ at roughly flat ranking (nDCG@10 of $0.332$) and a small ROI cost, for a balance of $0.411$, marginally below the baseline. The two levers then diverge sharply. The model-side margin loss pushes PC@10 to $0.918$ but spends accuracy to do so, dropping nDCG@10 to $0.309$. Its balance of $0.407$ sits below the baseline, so it buys coherence at the cost of the overall balance. The data-side regrouping instead lifts PC@10 to $0.949$ while *also* raising nDCG@10 to $0.352$ and leaving ROI@10 essentially unchanged. Its balance of $0.452$ is the highest of any configuration and the only one above the LightGCN baseline. Combining both levers reaches the highest raw PC@10 ($0.982$) but its balance ($0.444$) falls back below regrouping alone, because the margin loss spends accuracy even on the corrected bands. The headline is therefore that of the two exclusive levers, only regrouping raises the harmonic-mean balance.

### 6.3 Per-Band Suitability

The per-band lifts in Table 4 trace where each lever spends its effort. The margin loss is largely spent rescuing the two

| Configuration | nDCG@10 | ROI@10 | PC@10 | Balance | PC-lift@10 |
|---|---|---|---|---|---|
| Random Forest baseline | 0.019 | +0.014 | 0.667 | 0.054 | 1.06× |
| LightGCN baseline | 0.330 | −0.005 | 0.794 | 0.428 | 1.17× |
| Segmentation | 0.332 | −0.007 | 0.821 | 0.411 | 1.19× |
| Segmentation + margin (model lever) | 0.309 | −0.007 | 0.918 | 0.407 | 1.40× |
| Segmentation + regrouping (data lever) | 0.352 | −0.006 | 0.949 | **0.452** | n/a |
| Segmentation + regrouping + margin (combined) | 0.333 | −0.005 | 0.982 | 0.444 | n/a |

Table 3: Aggregate metrics averaged over the 69 temporal splits. ROI@10 is per month. Balance is the harmonic mean of the three axes with ROI rescaled (Section 5.2). PC-lift@10 = PC@10/$\pi(b)$ and is n/a for the regrouped configurations, whose shifted band mix makes aggregate lift incomparable. Regrouping is the only intervention whose balance exceeds the LightGCN baseline (0.452 vs 0.428).

| Band | $\pi(b)$ | RF | LightGCN | Segment. | +Margin | +Regroup | Combined |
|---|---|---|---|---|---|---|---|
| Conservative | 0.65 | 0.52 | 0.75 | 0.91 | 1.19 | 1.21 | 1.25 |
| Income | 0.78 | 0.70 | 1.13 | 1.20 | 1.26 | 1.25 | 1.27 |
| Balanced | 0.76 | 1.03 | 1.17 | 1.22 | 1.28 | 1.24 | 1.30 |
| Aggressive | 0.35 | 1.88 | 1.35 | 1.17 | 1.97 | 2.24 | 2.68 |

Table 4: Per-band PC-lift@10 (PC@10/$\pi(b)$) for the baselines and the segmentation configurations. Values $> 1$ beat the band-conditional random baseline $\pi(b)$. Both baselines under-serve the Conservative band, and regrouping is the first configuration to clear the random baseline in every band.

extremes that the baselines failed: it drags Conservative from 0.75 past the random baseline to 1.19 and lifts Aggressive from 1.35 to 1.97. Segmentation alone moves Conservative only to 0.91 (still below random) and even lowers Aggressive to 1.17, likely because the smaller per-band training graph yields weaker embeddings in isolation. Regrouping clears the random baseline in every band (1.21, 1.25, 1.24, 2.24), and adding the margin on top of it reaches the highest per-band lifts of all (1.25, 1.27, 1.30, 2.68). The Conservative deficit that both baselines exhibited is thus closed by either lever, but only regrouping closes it without the accuracy penalty seen in the aggregate.

### 6.4 Consistency Across Splits

The aggregate gains are not artefacts of averaging over a few strong splits. Figure 6 shows the per-split deltas against the baseline: PC@10 is positive on every one of the 69 splits for all three configurations, and the balance delta is positive for regrouping (+0.019 on average) but negative for both segmentation alone and the margin loss. Figure 7 reports the win rate, the fraction of splits on which a configuration beats the baseline. Regrouping wins the balance on 71% of splits and nDCG@10 on all of them, whereas the margin loss wins the balance on only 26% of splits and nDCG@10 on a mere 7%: its accuracy cost is a uniform downward shift, not a handful of bad splits. The per-split picture confirms the aggregate ranking of the levers.

### 6.5 Why Regrouping Beats the Margin Loss

The two levers attack the same problem from opposite ends, and the balance shows the difference. The margin loss fights the noisy declared labels during training by penalising discordant scores. It raises coherence but spends ranking accuracy doing so, and the per-band table shows that cost is largely incurred rescuing a single band. Regrouping instead corrects the labels at their source, routing and training each customer on the band implied by their revealed purchases, so coherence rises without any fight against the data. The decisive evidence is that nDCG@10 and ROI@10, both independent of the risk band, improve under regrouping: a model cannot rank held-out purchases better by relabelling bands unless the original labels were injecting noise into training. Their simultaneous improvement is direct evidence that the declared bands, not the model, were the binding constraint. The recommended operating point is therefore the data lever alone: once the bands are corrected, the margin loss is redundant, lowering the balance from 0.452 to 0.444 while adding only raw coherence the application does not need.

## 7 Limitations and Future Work

Beyond the data caveats already noted for the analytical study (Section 3.6), several limitations bound the experimental results, each pointing to a direction for future work.

### 7.1 Limitations

- *Single operating point and backbone.* We audit each baseline only at its primary-metric optimum (nDCG@10 for LightGCN, ROI@10 for Random Forest), and we report a single operating point for the margin lever on the inherited best-trial backbone hyperparameters rather than re-tuning end to end. The segmentation itself extends only LightGCN. The reported suitability gains are therefore a lower bound on what a full frontier sweep over sequential, transformer, or content-based recommenders could reach, and the accuracy costs are an upper bound.
- *Baseline-relative reading.* Per-band PC-lift@10 should be read alongside its $\pi(b)$ reference rather than in isolation, since low-$\pi(b)$ bands such as Aggressive clear the random bar more easily.

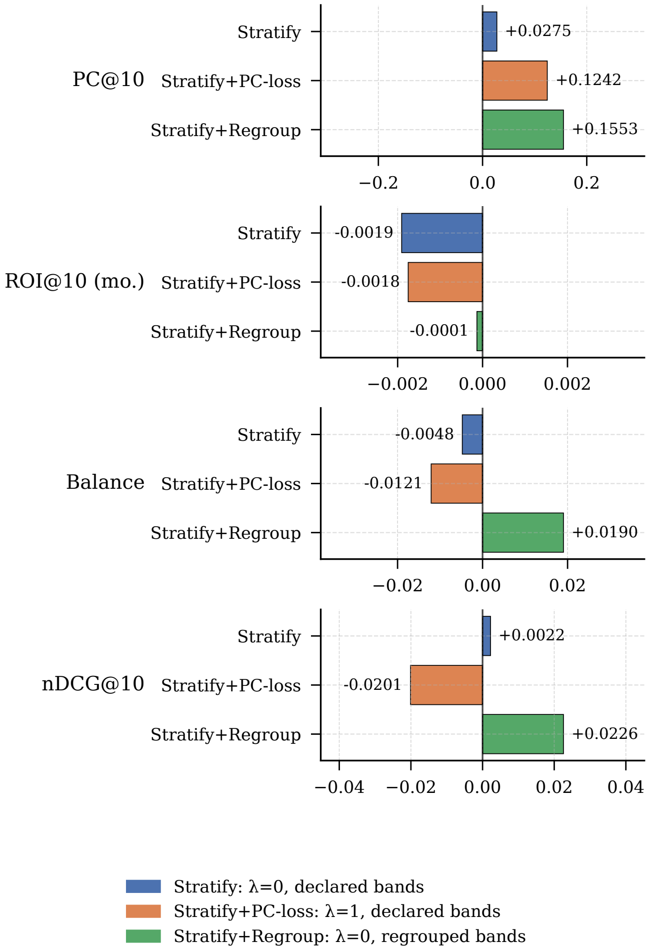


Figure 6: Paired per-split deltas against the LightGCN baseline for segmentation, the model lever, and the data lever. PC@10 is positive on every split, but only regrouping (data lever) yields a positive Balance delta.

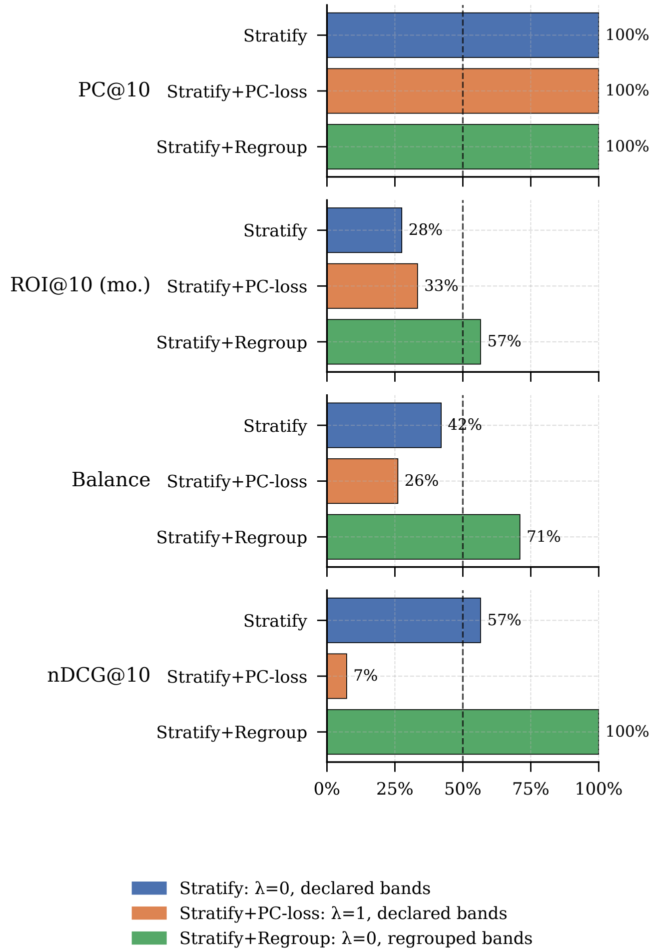


Figure 7: Per-split win rates against the LightGCN baseline. Regrouping wins the Balance on 71% of splits and nDCG@10 on all of them, while the margin loss wins the Balance on 26% and nDCG@10 on only 7%.

- *Regrouping leakage and tautology.* The revealed bands are derived from each customer's full Buy history, so they carry look-ahead information relative to each split. A per-split variant using only training-window Buys is left to future work. Because PC@10 is then scored against bands chosen to match those purchases, the regrouped suitability gain is partly guaranteed by construction. The trustworthy signal is the simultaneous nDCG@10 and ROI@10 improvement, neither of which is defined relative to the band, and which the balance metric captures.

## 7.2 Future Work

- *Causal identification.* The coherence premium of Section 3 is associational. Instrumental-variable or difference-in-differences designs on data spanning a regulatory boundary would move it towards causation, and would let the elevated 2018 mean discordance be tested directly against a pre-MiFID II counterfactual rather than left as a hypothesised transition effect.
- *Data-driven metric calibration.* The $\pm 1$ tolerance and the hierarchical asset-band assignment are set rather than derived. Future work could calibrate the tolerance from data and validate the band assignment against external MiFID II classifications from fund providers.
- *Broader backbones and a full frontier.* The segmentation and its two levers could be applied to sequential, transformer, and content-based recommenders, and the single margin operating point expanded into a full Pareto sweep with end-to-end re-tuning, mapping the balance frontier rather than reporting a single point on it.
- *Generalisation and online evaluation.* Results are shown on FAR-Trans under MiFID II's four-band scale. Extending to other jurisdictions, horizons, or asset universes would test the generalisability of PC@10 and the balance. Our analytical study measures realised return on executed Buys, whereas the experiments measure forward return on top-10 slates that may never be acted on. Bridging this gap through online evaluation or counterfactual off-policy estimation is a promising direction.

## 8 Conclusion

The central finding of this paper is that return, ranking accuracy, and regulatory risk suitability are not in structural conflict on FAR-Trans: a recommender can be made substantially more suitable to the customer's declared risk band while keeping, and even improving, the accuracy and return the data already supports. This reframes suitability as an architectural choice rather than a tradeoff to be managed by post-hoc reranking, expressible as an ordinal distance on a customer-asset band pair and optimisable inside the model.

The results assemble a diagnostic-to-remedy chain. An analytical study establishes that discordance is prevalent and a persistent customer-level trait, and that coherent Buys earn a statistically significant return premium, removing the economic objection to optimising for suitability. Auditing the FAR-Trans baselines on the three axes shows that both underserve specific risk bands and that neither balances the three. Risk-based segmentation then closes the per-band deficit. Of its two exclusive levers, regrouping customers onto their revealed band is the dominant one: it lifts the harmonic-mean balance above the baseline by raising suitability and accuracy together at no cost to return. The margin loss, by contrast, buys suitability by spending accuracy, and so lowers the balance. The measurement apparatus that makes this visible, Profile Coherence at $k$ and the harmonic-mean balance, is a secondary contribution.

More broadly, the approach shows that regulatory suitability constraints can be encoded as training-time objectives and, better still, corrected at the data source rather than filtered afterwards. This opens a path for recommenders in other regulated domains, such as insurance product suitability and credit risk-appetite matching, to embed compliance directly into their optimisation.

## 9 Reproducibility

All code, configuration grids, and evaluation artefacts are available in the project repository at https://github.com/Samthesimpsons/SMU-Capstone.

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